

Yu-Shiang Huang

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Nationality: Taiwan (R.O.C.)

Date of Birth: 04 November 1997

Summary

I am a Ph.D. candidate in Data Science at National Taiwan University. My research interests include retrieval-enhanced machine learning (REML), natural language processing (NLP), and recommender systems (RecSys), with a particular focus on their applications in the financial domain.

Education

Ph.D.	National Taiwan University Data Science	Since 2022
M.S.	National Taiwan University Data Science	Admitted to the Ph.D. program in September 2022
B.S.	National Taiwan University Accounting	2020

Positions

National Institute of Advanced Industrial Science and Technology (AIST) Visiting Intern
Tokyo, Japan January 2025-March 2025

- Decision-oriented Text Evaluation: I worked with Chung-Chi Chen to develop a decision-oriented text evaluation method, which focuses on finding an alternative evaluation method for market commentary generation.

Academia Sinica Graduate Student Researcher
Taipei, Taiwan September 2020-present

- REML Research: Designed task formulation and an end-to-end retrieval-augmented pipeline to extract important signals from financial reports.
- RecSys Research: Designed experiments to validate the effectiveness of a new loss function compared to state-of-the-art (SOTA) recommendation methods.
- RecSys Industry Cooperation: Researched multi-behavior algorithms for banner recommendation systems and deployed the developed model in a real-world system with millions of users.
- NLP Research: Analyzed the characteristics of financial annual reports and designed a new algorithm to classify sentence relations based on these findings. Leveraging the results, we conducted transfer learning in a self-supervised setting, and our paper was accepted by ACL 2023.
- NLP Industry Cooperation: Fine-tuned BERT with a Siamese network and contrastive loss to obtain sentence embeddings in legal data, building a retrieval system aimed at finding the most relevant internal regulations.

National Taiwan University
Taipei, Taiwan

Head Teaching Assistant, CSIE5043 Machine Learning
September 2023-January 2024

- Teaching Assistance: Led a team of 10 to assist in lecturing Machine Learning at National Taiwan University.
- Project Design: Designed the final project and maintained the data streaming pipeline.
- Homework Validation: Validated the homework problem sets related to machine learning theory.

National Taiwan University
Taipei, Taiwan

Teaching Assistant, CSIE5043 Machine Learning
September 2021-January 2022

- Final Project Design: Created an in-class competition for a 300-student course by adding noise to an open dataset and testing competition metrics with various machine learning models.
- Assignment Validation and Consulting: Validated assignments to ensure accuracy in problem sets and provided answers to machine learning theory-related questions during TA hours.

National Taiwan University
Taipei, Taiwan

Teaching Assistant, GENEDU5010 Programming and Web Scraping
February 2021-June 2021

Instructional Support: Possessed a background in designing programming assignments and offering instructional support, allowing me to contribute effectively to teaching endeavors.

beBit Inc.
Taipei, Taiwan

Part-time Quantitative Analyst
October 2020-March 2021

- Survey Data Analysis: Collaborated with a qualitative research team and applied statistical methods to verify business hypotheses and identify key factors in the customer experience journey.
- Methodology Establishment: Improved quantitative analysis methodology based on research papers and technical blogs, enhancing the persuasiveness of the final deliverable.

BluePlanet Technology
Taipei, Taiwan

Project Assistant
November 2019-February 2021

- Product Mechanism Design: Surveyed literature related to accounting and regulation to support the product mechanism design, resulting in the successful launch of the product in January 2021.
- Communication: Bridged the gap between the finance research group and the product management team by facilitating communication and recording discussion details during meetings.

National Taiwan University
Taipei, Taiwan

Part-time Research Assistant
July 2020-September 2020

- Website Development at Department of Economics: Utilized Python to create a website for an economics experiment based on a research paper in experimental economics, catering to a participant pool of 100 individuals.

Academia Sinica
Taipei, Taiwan

Intern
July 2019-June 2020

- Publication: Completed and submitted the work "NFinBERT: A Number-Aware Language

Model for Financial Disclosures” to SwissText 2021, which has been accepted.

- Textual Data Analysis: Implemented machine learning models to conduct experiments on the relationship between financial report text and financial measures.
- Experimental Data Preparation: Collected SEC 10-K filings from 1993 to 2018 and performed data cleansing on the corpus through shell scripting, making the 330 GB data available for further research.

National Taiwan University
Taipei, Taiwan

Teaching Material Assistant
November 2018-January 2019

- Teaching Material Creation at Department of Electronic Engineering: Self-studied advanced topics about kernel tricks in Support Vector Machines (SVM) in machine learning. Developed teaching slides with explainable examples beyond textbooks, aimed at making the theorems easier to understand.

Publications

Conference Papers

Yu-Hsiang Wang, Wei-Ning Chiu, Yi-Tai Hsiao, Yu-Shiang Huang, Yi-Shyuan Chiang, Shuo-En Wu, and Chuan-Ju Wang. SURF: A system to unveil explainable risk relations between firms. In *2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics – System Demonstration Track*, 2025

Shao-En Lin, Brian Liu, Miao-Chen Chiang, Ming-Yi Hong, Yu-Shiang Huang, Chuan-Ju Wang, and Che Lin. Betag: Behavior-enhanced item tagging with finetuned large language models. In *THE WEB CONFERENCE 2025*

Jia-Huei Ju, Yu-Shiang Huang, Cheng-Wei Lin, Che Lin, and Chuan-Ju Wang. A compare-and-contrast multistage pipeline for uncovering financial signals in financial reports. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14307–14321, 2023

Ta-Wei Huang, Jia-Huei Ju, Yu-Shiang Huang, Cheng-Wei Lin, Yi-Shyuan Chiang, and Chuan-Ju Wang. FISH: A Financial Interactive System for Signal Highlighting. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 50–56, 2023

Hao-Lun Lin, Jr-Shian Wu, Yu-Shiang Huang, Ming-Feng Tsai, and Chuan-Ju Wang. NFinBERT: A Number-Aware Language Model for Financial Disclosures (short paper). In *SwissText*, 2021

Workshop Papers

Yu-Shiang Huang, Yu-Hsiang Wang, Yi-Tai Hsiao, Che Lin, and Chuang-Ju Wang. Finlight: Bridging the annotation gap for highlight extraction in financial reports. Presented at FinIR 2025: The 2nd Workshop on Financial Information Retrieval in the Era of Generative AI, co-located with the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, July 2025. Non-archival workshop presentation

Wei-Ning Chiu, Yu-Hsiang Wang, Yi-Tai Hsiao, Yu-Shiang Huang, and Chuan-Ju Wang. Financial risk identification through dual-view adaptation. Presented at FinIR 2025: The 2nd Workshop on Financial Information Retrieval in the Era of Generative AI, co-located with the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, July 2025. Non-archival workshop presentation

Recognition**Awards**

Legal Compliance Special Award, Lawsnote Legal-Tech Hackathon, 2021

Service**Organizational Leadership**

ccClub (a Python Community), Director and Teaching Assistant, 2018-2024

Skills

- Extensive programming experience with Python, particularly for machine learning projects on Unix platforms.
- Proficient in machine learning related packages such as PyTorch, Transformers, Sentence-BERT, LangChain, Scikit-learn, NumPy, Pandas and Matplotlib.
- Fluent in English and native proficiency in Traditional Chinese.